

Supplementary Material for ContextLens

Technical Architecture and Data Flow

Anonymous Submission

Anonymous artifacts. An anonymous repository containing the complete source code and demonstration video is available at <https://anonymous.4open.science/r/ContextLens-B7C0/README.md>.

Supplement A. Software Architecture

ContextLens separates interaction, evidence compilation, data services, and presentation (Figure A1). A local controller receives a dated address and maintains the investigation state. The evidence compiler resolves temporal place identity before retrieval. It then normalizes records, admits relations and claims, and projects the audited objects into a dossier.

The packaged snapshot and live adapters share a normalized record contract. Source Passports retain the evidence identifier, original URI, temporal scope, and normalization state. Model-assisted interpretation occurs after the claim audit. Returned identifiers are intersected with the admitted set before a summary is exposed. Model failure therefore affects the optional summary, not the underlying dossier.

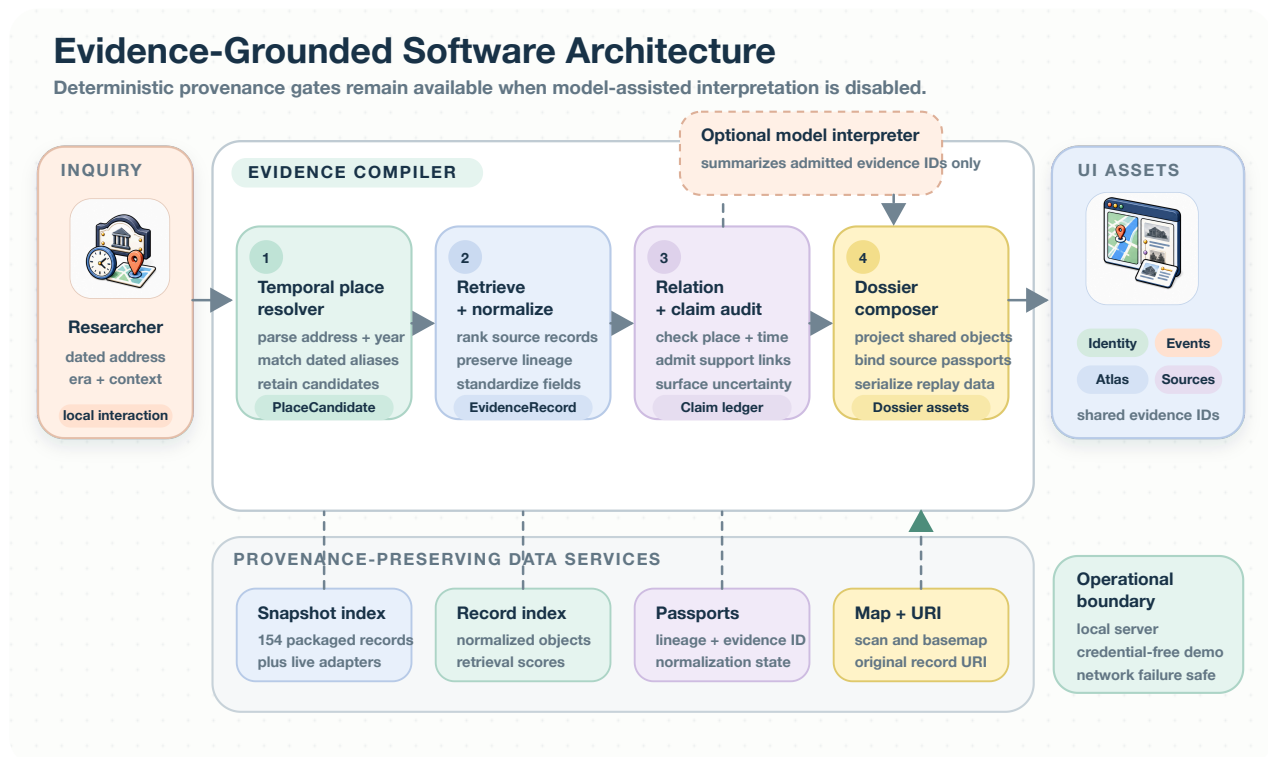


Figure A1: Software architecture. The deterministic evidence compiler mediates between a dated inquiry and four interface views. The optional model interpreter can summarize only evidence that has passed relation and claim gates. Dashed connections denote data-service or optional dependencies.

Supplement B. Data Flow Pipeline

The pipeline distinguishes operations from the artifacts they produce (Figure A2). Source records comprise the packaged official snapshot, catalog and map metadata, and open record URIs. Query Data contains the address string, temporal constraint, and optional contextual clue. Resolution and retrieval produce candidate place identities, ranked records, and the original source fields required for normalization.

Processing standardizes schemas, removes duplicates, and aligns temporal scopes. Its outputs are typed evidence objects, Source Passports, and linked entities. Analysis applies relation admission and claim auditing before optional model summarization. The renderer projects shared evidence objects into the Identity, Events, Atlas, and Sources views and the dossier JSON.

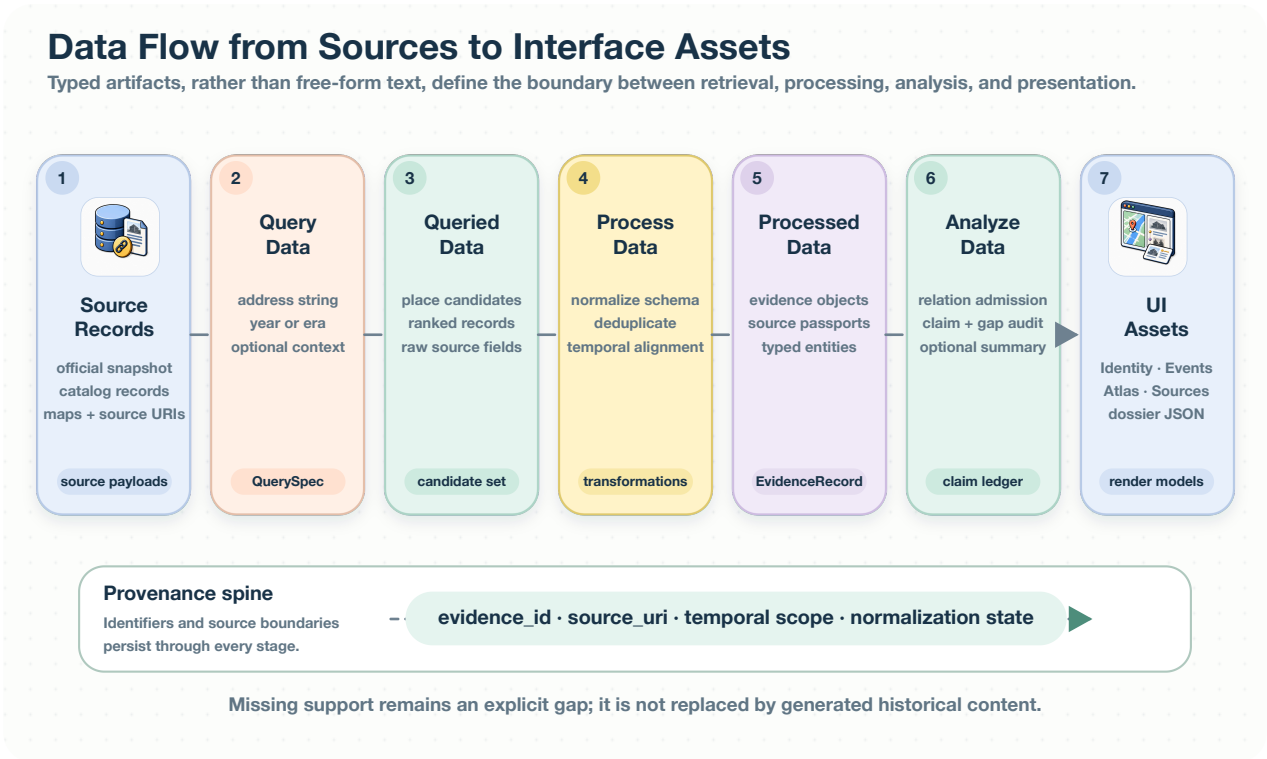


Figure A2: Data flow from source records to interface assets. Each stage produces a named artifact, while the provenance spine preserves the evidence identifier, source URI, temporal scope, and normalization state across transformations.